

TECHNICAL REPORT

Predicting Health Insurance Charges from Demographic and Lifestyle Factors

A Regression Analysis of Age, BMI, Smoking Status, and Related Factors Across 1,338 Policyholders

Tools: Python · pandas · NumPy · Matplotlib · Seaborn · scikit-learn

Notebook: insurance.ipynb

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1. Introduction

1.1 Background

Health insurance providers price policies by estimating, in advance, how much a given policyholder is likely to cost them in medical claims. That estimate depends on a mix of factors that are only partially within a person's control — age and sex are fixed, while body mass index (BMI) and smoking status reflect lifestyle choices. Understanding how strongly each of these factors drives actual billed charges is valuable both to insurers setting premiums and to individuals trying to understand what is actually shaping their own healthcare costs.

This project uses a dataset of 1,338 insurance policyholders to answer a practical question: **which personal and lifestyle factors most strongly predict medical insurance charges, and how accurately can those charges be predicted using demographic and behavioural data alone?**

1.2 Why This Project Matters

Beyond the specific numbers produced, this project has value on three levels.

- **Practically**, identifying which factors most strongly drive insurance charges mirrors exactly the kind of analysis an actuarial or pricing team would run before setting premium tiers — knowing that one factor dominates the cost structure, for example, changes how a pricing model should be built.
- **Analytically**, this dataset provides a genuine second test of the workflow built in an earlier project (student academic performance), but under different conditions — no missing data, entirely nominal categorical variables instead of a mix of ordinal and nominal, and a target with a real non-linear interaction effect. Comparing how the same pipeline performs under these different conditions is itself a valuable exercise in judgement, not just code reuse.
- **Developmentally**, this project was carried out independently as deliberate practice: applying a previously learned regression workflow to a new, unfamiliar dataset without external reference code, in order to test whether the underlying concepts — encoding strategy, scaling, train/test splitting, model comparison, and diagnostic checking — had actually been internalised rather than simply copied.

1.3 Objectives

The project set out to:

1. Load, inspect and understand a real-world medical insurance dataset covering demographic, lifestyle and cost variables.

2. Explore the distribution of every variable, both categorical and numerical, before any modelling took place.
3. Convert all categorical variables into a numeric form appropriate to their nature — binary mapping for two-category variables, one-hot encoding for the unordered, multi-category region variable.
4. Quantify the relationship between each factor and insurance charges using correlation analysis and targeted visualisations.
5. Build and compare four regression models to predict charges, and diagnose which one is best suited to the true structure of the data.
6. Verify the chosen model with overfitting checks, cross-validation, and residual analysis, rather than relying on a single R^2 score in isolation.

1.4 Scope and Approach

The analysis follows the same supervised machine-learning workflow used in prior work: data acquisition → data quality assessment → exploratory data analysis → feature encoding → feature scaling → train/test split → model training → model comparison → diagnostic verification → interpretation.

The target variable is *charges*, a continuous numeric value representing the amount billed by the health insurer. Because the target is continuous, this is a **regression** task, which determines the algorithms used and the metrics reported (R^2 and RMSE, rather than classification metrics such as accuracy).

1.5 Tools and Libraries

Library	Role in this project
pandas	Loading the CSV, inspecting structure, encoding categorical variables, and all DataFrame manipulation
NumPy	Underlying numerical array operations
Matplotlib	Histogram grids and the residual plot
Seaborn	Correlation heatmap, boxplot, scatter plot, and pairplot
scikit-learn	StandardScaler, train_test_split, LinearRegression, Ridge, Lasso, RandomForestRegressor, cross_val_score, and the r2_score / mean_squared_error metrics

1.6 A Note on Independent Work

This project was completed independently, without referencing any external notebook or third-party code for this specific dataset. It reuses a general regression workflow developed and documented in an earlier project on a different dataset (student academic performance), applied here as a deliberate second exercise

to test and reinforce the same underlying concepts — categorical encoding strategy, feature scaling, train/test methodology, and model comparison — against a new problem with a different structure. All code, chart interpretation, and analytical conclusions in this report are the author's own.

2. Story of the Data

2.1 Source and Structure

The dataset, `insurance.csv`, contains **1,338 records and 7 columns**. Each row represents one primary beneficiary — the main policyholder registered under an insurance plan, as distinct from any dependents covered under the same policy.

Confirmed via `df.info()`:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

2.2 Data Dictionary

#	Column	Type	Description
0	age	int64	Age of the primary beneficiary (18–64)
1	sex	object	Female / Male
2	bmi	float64	Body mass index — weight relative to height
3	children	int64	Number of dependents covered by the policy
4	smoker	object	Yes / No
5	region	object	Residential region: northeast / northwest / southeast / southwest
6	charges	float64	Target variable — individual medical costs billed by the insurer

2.3 Data Quality — No Missing Values

Unlike the student habits dataset used in earlier work, this dataset required **no imputation**. Every one of the 7 columns returned a non-null count of 1,338 in `df.info()`, meaning the dataset arrived complete. This simplified the preprocessing stage considerably: there was no missing-value strategy to design or justify, and the analysis could proceed directly from inspection to encoding.

2.4 Summary Statistics

Statistic	age	bmi	children	charges
count	1338	1338	1338	1338
mean	39.21	30.66	1.09	13,270.42
std	14.05	6.10	1.21	12,110.01
min	18	15.96	0	1,121.87
25%	27	26.30	0	4,740.29
50%	39	30.40	1	9,382.03
75%	51	34.69	2	16,639.91
max	64	53.13	5	63,770.43

Two things stand out immediately. First, `charges` has a very wide spread relative to its mean — a standard deviation ($\approx 12,110$) nearly as large as the mean itself ($\approx 13,270$) — which is an early signal that this variable is right-skewed rather than symmetric, confirmed later in Section 4. Second, `age` spans a realistic adult working-age range (18 to 64), with 64 sitting just under the U.S. Medicare eligibility age of 65, beyond which primary health coverage typically shifts away from private insurance.

2.5 Categorical Variable Summary

df[newlist].describe()			
	sex	smoker	region
count	1338	1338	1338
unique	2	2	4
top	male	no	southeast
freq	676	1064	364

The sample is close to evenly split by sex (676 male out of 1,338, $\approx 50.5\%$). The great majority of policyholders are non-smokers (**1,064 of 1,338, or 79.5%**), meaning smokers make up a minority group of roughly 20.5% — a class imbalance worth keeping in mind when interpreting the strength of any smoker-related effect. Region is fairly evenly distributed across the four categories, with southeast the most common at 364 records.

3. Data Splitting and Preprocessing

3.1 Separating Column Types

```
newlist = df.select_dtypes(include='object').columns
newlist = [item for item in newlist]
```

`select_dtypes(include='object')` isolates the text columns — sex, smoker, region — and the list comprehension converts the resulting pandas Index into a plain Python list so it can be indexed later (e.g. `newlist[g]` in the plotting loop). Unlike the earlier student project, there is no identifier column to remove here — this dataset has no equivalent of `student_id`, so every column carries genuine information.

```
newlist2 = df.select_dtypes(exclude='object').columns
newlist2 = [item for item in newlist2]
```

The complementary operation isolates the four numerical columns: age, bmi, children, and charges (the target).

3.2 Encoding Strategy

A key structural difference from the earlier student project: every categorical column in this dataset is nominal — none of them are ordinal. There is no natural ranking to Female vs Male, Yes vs No (smoker), or any of the four regions. This means the ordinal `.map()`-with-ranked-dictionary technique used for variables like diet quality in the earlier project does not apply here in its ranking sense — though the mechanical use of `.map()` is still the right tool for the two binary columns, simply without implying any order.

3.2.1 Binary Encoding — sex and smoker

Both sex and smoker have exactly two categories, so a single 0/1 column captures all the available information — there is no need for a full one-hot expansion, since a two-category one-hot encoding with `drop_first=True` reduces to exactly this.

```
df2['sex_e'] = df2['sex'].map({'female': 0, 'male': 1})
df2['smoker_e'] = df2['smoker'].map({'no': 0, 'yes': 1})
```

The 0 and 1 here are **labels, not a rank** — there is no implication that "male" is greater than "female." With only two categories, this distinction is safe: a model cannot mistake a binary label for an ordered scale the way it could with three or more unranked categories.

3.2.2 One-Hot Encoding — region

Region has four unordered categories, so assigning arbitrary numbers (0, 1, 2, 3) would inject a false hierarchy — for example implying "southwest" is worth three units more than "northeast" for no real reason. One-hot encoding avoids this by giving each category its own binary column:

```
region_dummies = pd.get_dummies(df2['region'], prefix='region', drop_first=True)
df3 = pd.concat([df2, region_dummies], axis=1)
```

`drop_first=True` removes one category (northeast, alphabetically first) as the baseline, avoiding redundancy: if a row shows 0 across `region_northwest`, `region_southeast`, and `region_southwest`, it must be northeast. This produced 12 columns in `df3` before cleanup.

3.2.3 Removing the Original Text Columns

```
df3 = df3.drop(['sex', 'smoker', 'region'], axis=1)
```

With numeric equivalents now in place for all three categorical variables, the original text columns were dropped, leaving a fully numeric, 9-column DataFrame ready for correlation analysis and modelling.

3.3 Defining Features and Target

```
df4 = df3
x = df4.drop('charges', axis=1)
y = df4['charges']
x.shape, y.shape  # ((1338, 8), (1338,))
```

`charges` is the value being predicted, so it is excluded from `x` (the eight predictors: `age`, `bmi`, `children`, `sex_e`, `smoker_e`, and three region dummies) and isolated into `y`. This mirrors the exact reasoning used for `exam_score` in the earlier project: the target can never appear among its own predictors, or the model would effectively be handed the answer.

3.4 Feature Scaling

```
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
x_scaled.shape  # (1338, 8)
```

Every feature is rescaled to mean 0 and standard deviation 1, preventing `age` (ranging up to 64) from numerically dominating a binary column like `smoker_e` purely because of its larger scale, before either has been evaluated for actual importance.

3.5 Train / Test Split

```
x_train, x_test, y_train, y_test = train_test_split(
    x_scaled, y, test_size=0.2, random_state=42
)
# ((1070, 8), (268, 8), (1070,), (268,))
```

The 1,338 records were split into **1,070 training rows (80%)** and **268 test rows (20%)**, with `random_state=42` fixing the split for reproducibility — the identical convention used throughout the earlier project.

4. Pre-Analysis

Before any modelling, every variable's shape was inspected — not to draw conclusions yet, but to catch anything (extreme skew, an impossible value, a badly imbalanced category) that would need addressing before the model stage.

4.1 Categorical Variable Distributions

Figure 1 (Section 7) shows histograms for sex, smoker, and region. Sex is close to balanced. Smoker shows a pronounced imbalance — non-smokers substantially outnumber smokers, consistent with the 79.5% figure from Section 2.5. Region is close to evenly spread across all four categories, with no single region dominating.

4.2 Numerical Variable Distributions

Figure 2 (Section 7) shows histograms for the four numerical columns:

- age is close to uniformly distributed across the 18–64 range, with one clear exception: a pronounced spike at the youngest age (18–19). This likely reflects a genuine feature of insurance enrolment data — a large cohort of policyholders signing up at the minimum eligible age — rather than a data error.
- bmi is the most conventionally bell-shaped of the four variables, approximately symmetric and centred around the high-20s to low-30s range, consistent with its mean of 30.66 in the summary statistics.
- children is a discrete count variable, heavily right-skewed: the majority of policyholders have zero dependents, with rapidly shrinking counts at 1, 2, 3, and very few at 4 or 5.
- charges — the target — is strongly right-skewed. The bulk of policyholders are billed under roughly \$15,000, with a long thinning tail extending out toward \$60,000+. This is the shape typical of cost and income data generally: most people cluster at moderate values, while a minority incur very large costs.

4.3 Why the Skew in Charges Matters

Because charges is right-skewed rather than normally distributed, this is worth flagging as a modelling consideration: linear regression does not strictly require a normally distributed target, but it does assume that prediction errors (residuals) are roughly normally distributed and of consistent size across the range of predictions. A skewed target is often — though not always — an early warning sign of exactly the kind of residual pattern uncovered later in Section 5.3. In some regression workflows this would motivate a log-transformation of the target before modelling; that step was not applied here, and its effect is a natural direction for future refinement (see Section 8.3).

5. In-Analysis

5.1 Correlation Analysis

```
corr = df3.corr()
sns.heatmap(corr, annot=True)
```

With every column now numeric, the full correlation matrix (Figure 3, Section 7) reveals each factor's linear relationship with charges:

Feature	Correlation with charges	Interpretation
smoker_e	+0.79	Very strong positive — the dominant predictor by a wide margin
age	+0.30	Moderate positive
bmi	+0.20	Weak-to-moderate positive
region_southeast	+0.07	Very weak positive
sex_e	+0.06	Negligible
children	+0.07	Negligible
region_northwest	−0.04	Negligible
region_southwest	−0.04	Negligible

smoker_e dominates every other feature by a wide margin. At +0.79, its correlation with charges is more than two and a half times stronger than the next-best predictor (age, +0.30), and nearly four times stronger than bmi. This single binary variable carries the overwhelming majority of the linear signal in the dataset.

A secondary detail worth noting from the same heatmap: `bmi` and `region_southeast` show a correlation of +0.27 with each other — policyholders in the southeast region trend toward higher BMI than other regions. This is a relationship between two *predictors* rather than with the target, and does not affect the regression directly, but it is a genuine pattern in the data worth being aware of.

5.2 Baseline Model — Linear Regression

```
lr_model = LinearRegression()
lr_model.fit(x_train, y_train)
lr_model.score(x_test, y_test)
```

Result: $R^2 = 0.7836$

The baseline linear model explains approximately **78.4% of the variance** in insurance charges on the held-out test set. This is a solid result, though noticeably lower than the $\approx 89\%$ achieved on the earlier student-performance dataset — a difference that becomes clearer over the rest of this section.

5.3 Model Comparison

```
models = {
    "Linear": LinearRegression(),
    "Ridge": Ridge(alpha=1.0),
    "Lasso": Lasso(alpha=0.1),
    "Random Forest": RandomForestRegressor(n_estimators=100, max_depth=5,
random_state=42)
}

for name, model in models.items():
    model.fit(x_train, y_train)
    y_pred = model.predict(x_test)
    rmse = mean_squared_error(y_test, y_pred)
    print(f"{name} -> R²: {r2_score(y_test, y_pred):.3f}, RMSE: {rmse:.2f}")
```

A technical note on this cell's output: the variable is labelled `rmse`, but it is computed using `mean_squared_error(y_test, y_pred)` without taking the square root. What actually printed is therefore **MSE (mean squared error)**, not RMSE — the true root-mean-squared error is the square root of the printed figure. Both the raw printed values and the corrected RMSE are shown below for transparency.

Model	R ²	Printed value (MSE)	Corrected RMSE ($\sqrt{\text{MSE}}$)
Linear Regression	0.784	33,596,915.85	\$5,796.28
Ridge ($\alpha = 1.0$)	0.784	33,604,956.54	\$5,796.98
Lasso ($\alpha = 0.1$)	0.784	33,597,338.15	\$5,796.32
Random Forest	0.873	19,680,221.58	\$4,436.24

Read correctly, the **corrected RMSE** column is the genuinely interpretable figure: the linear-based models are typically off by about **\$5,796** per prediction, while Random Forest cuts that error down to about **\$4,436** — a meaningfully tighter, more useful prediction, consistent with its higher R^2 .

5.4 Overfitting Check

```
print("Train R²:", lr_model.score(x_train, y_train))
print("Test R²:", lr_model.score(x_test, y_test))

Train R²: 0.7417
Test R²: 0.7836
```

These two scores are close, which is the sign of a model that generalises rather than memorises. Notably, the **test score is actually slightly higher than the training score** here (0.784 vs 0.742) — the reverse of what overfitting would look like. This pattern is not unusual with a fixed random split on a moderately sized dataset; it simply reflects that this particular 20% test slice happened to be marginally easier to predict than the 80% training slice, and it is one reason cross-validation (Section 5.5) is a more reliable check than a single train/test comparison.

5.5 Cross-Validation

```
from sklearn.model_selection import cross_val_score
scores = cross_val_score(LinearRegression(), x_scaled, y, cv=5, scoring='r2')

[0.7615, 0.7065, 0.7781, 0.7327, 0.7556]
Mean: 0.7469
```

Across five different train/test splits, R^2 ranged from a low of **0.706** to a high of **0.778**, averaging **0.747**. This confirms that the single test-set score of 0.784 obtained earlier was reasonably representative, though slightly on the favourable end of the range — the more defensible figure to quote as "the" performance of the linear model is the cross-validated mean of approximately **0.75**, not the single-split figure of 0.784. This distinction matters: a report that only cites the best-looking split without cross-validating it risks overstating the model's reliability.

5.6 Residual Analysis

```
y_pred = lr_model.predict(x_test)
residuals = y_test - y_pred

plt.scatter(y_pred, residuals)
plt.axhline(0, color='red')
```

This is the most diagnostically important chart in the report (Figure 7, Section 7). For a linear model whose assumptions are well satisfied, residuals should scatter randomly and evenly around zero, with no visible structure. **That is not what this plot shows.** Instead, the residuals fall into distinct diagonal bands, and the spread of residuals visibly widens at higher predicted charges rather than staying constant. This is a clear, visual signature that a single straight-line relationship cannot fully capture how charges are generated — which is discussed fully in Section 6.

6. Post-Analysis and Insights

6.1 Smoking Status Is the Dominant Cost Driver

At a correlation of +0.79, `smoker_e` is not simply the strongest predictor — it stands in a different category altogether from every other variable in the dataset. The boxplot in Figure 4 makes this concrete: the **median charge for smokers (≈\$34,500) is roughly 4.5 times higher than the median for non-smokers (≈\$7,500)**, with almost no overlap between the two groups' main distributions. Smoking status alone explains the majority of the variation this model is able to capture.

6.2 The Real Story Is an Interaction, Not Just an Effect

The scatter plot of BMI against charges, coloured by smoking status (Figure 5), reveals something a correlation coefficient cannot: **smoking and BMI interact rather than simply adding together**. Among non-smokers, charges rise only gently with BMI across the entire range. Among smokers, charges also rise gently with BMI up to about 30 — but above a BMI of roughly 30, smokers' charges jump sharply into a visibly separate, much higher band, while non-smokers at the same BMI show no comparable jump.

In plain terms: **being a smoker is costly on its own, but being an obese smoker is disproportionately costly — far more than the sum of the two effects individually**. This is a classic statistical interaction effect, and it is precisely the kind of relationship a plain linear regression structurally cannot represent, because linear regression assumes each feature's effect is added independently rather than multiplied together under certain conditions.

6.3 This Explains Every Other Result in the Report

Once this interaction is understood, three otherwise-separate findings connect into a single, coherent explanation:

- **The residual pattern (Section 5.6)** — the banding and widening spread in the residual plot is the direct fingerprint of the model trying and failing to fit a single straight line through what is actually two different relationships (smoker and non-smoker) that only diverge sharply at high BMI.
- **Random Forest's advantage (Section 5.3)** — tree-based models split the data into regions and can naturally represent "if smoker AND bmi > 30, charges are very high" as a rule, which is exactly the structure a linear model cannot express. This is why Random Forest's R^2 (0.873) and RMSE (\$4,436) both meaningfully outperform the three linear-based models, mirroring — but for the opposite underlying reason — the pattern in the earlier student project, where the simplest model won because the true relationship actually was linear.

- **The right-skew in charges (Section 4.2)** — the long tail of very high charges in the histogram is now explainable: it is disproportionately made up of the smoker-plus-high-BMI subgroup, not a random collection of outliers.

6.4 Age Plays a Steady, Secondary Role

At +0.30, age is a real but modest predictor — consistent with the general expectation that healthcare costs rise gradually over a working lifetime, without the dramatic step-change seen with smoking. The pairplot (Figure 6) shows this as a gentle upward drift in charges with age within both the smoker and non-smoker groups, rather than any sharp threshold effect.

6.5 Sex, Children, and Region Carry Almost No Signal

Sex (+0.06), number of children (+0.07), and all three region dummies (ranging from -0.04 to $+0.07$) sit close enough to zero to be considered negligible linear predictors of charges in this dataset. This does not necessarily mean these factors are irrelevant in every insurance context — only that, within this particular sample, they carry far less explanatory power than smoking status, BMI, and age.

6.6 Overall Model Assessment

Using the more defensible cross-validated estimate, the linear model explains roughly **75% of the variance** in charges, while Random Forest reaches **87.3%** on the single held-out test set. The gap between these two numbers is not a sign that the linear model was built incorrectly — it is a direct, measurable consequence of the smoker×BMI interaction documented in Section 6.2, which by construction sits outside what any purely additive linear model can capture.

7. Data Visualisations and Charts

7.1 Figure 1 — Distribution of Categorical Variables

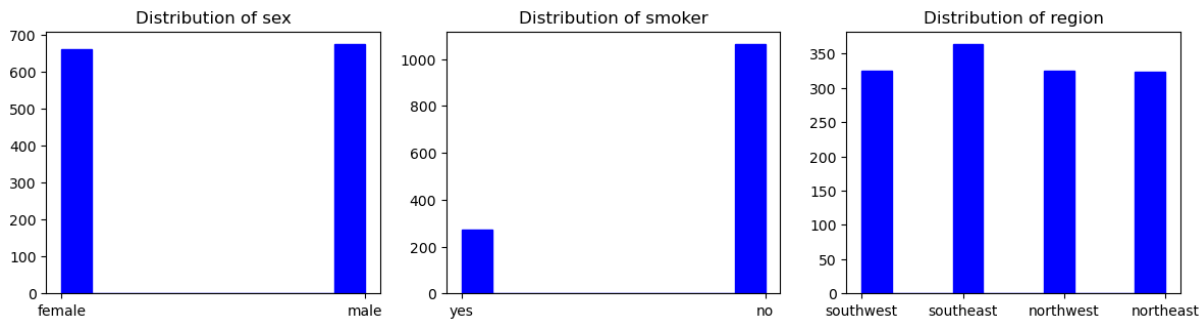


Figure 1. Distribution of sex, smoker, and region.

What it shows: a roughly balanced sex split, a pronounced imbalance in smoking status (non-smokers far outnumber smokers), and a fairly even spread across the four regions.

7.2 Figure 2 — Distribution of Numerical Variables

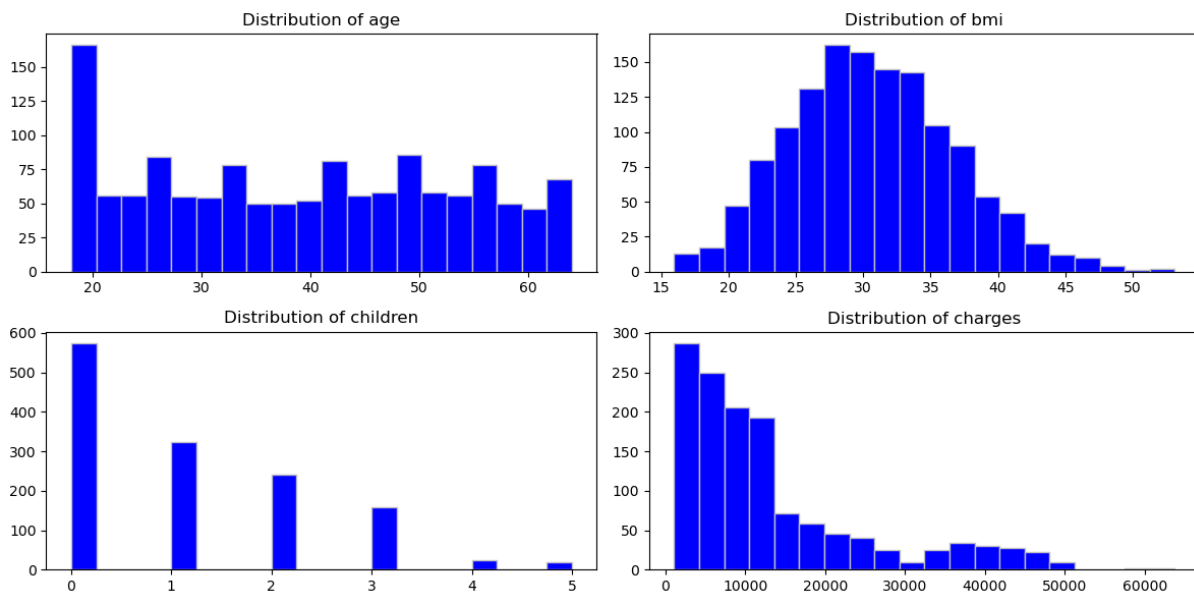


Figure 2. Distribution of age, bmi, children, and charges.

What it shows: a near-uniform age distribution with a spike at the minimum eligible age; an approximately normal BMI distribution; a heavily right-skewed children count; and a strongly right-skewed charges distribution, foreshadowing the residual pattern uncovered later.

7.3 Figure 3 — Correlation Heatmap

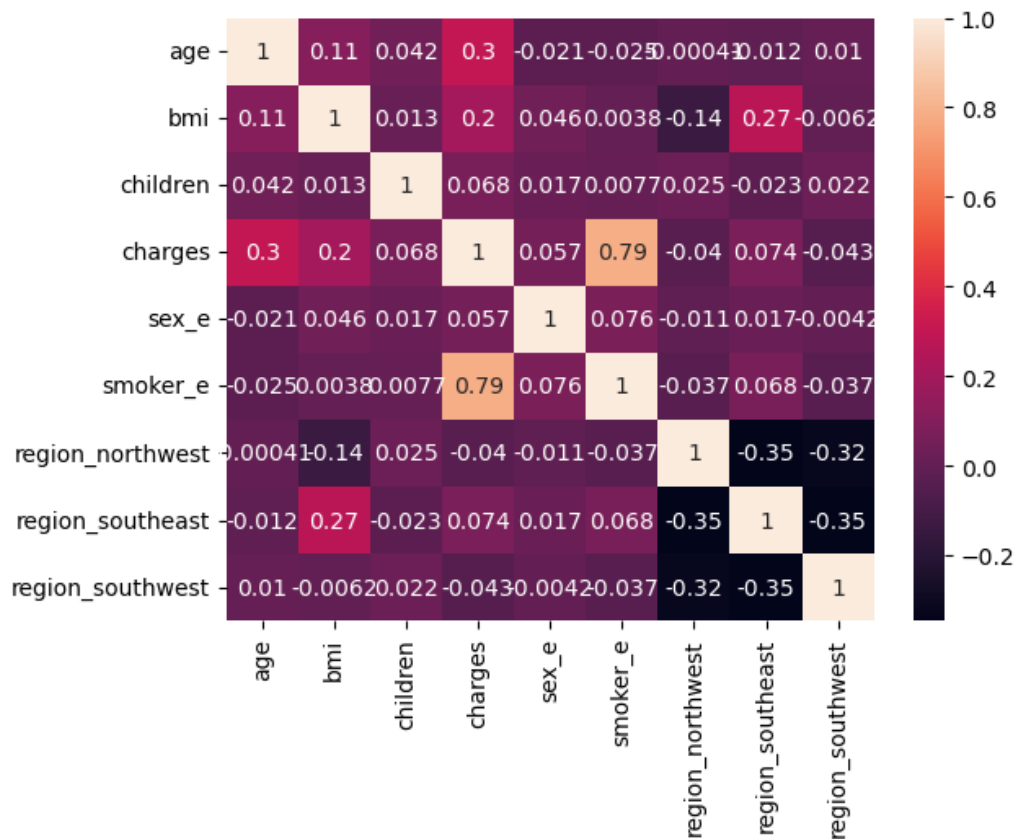


Figure 3. Correlation heatmap across all encoded variables.

What it shows: smoker_e stands out as by far the brightest cell in the charges row (+0.79), with every other feature comparatively muted — a direct visual confirmation of Section 6.1’s finding.

7.4 Figure 4 — Boxplot: Charges by Smoking Status

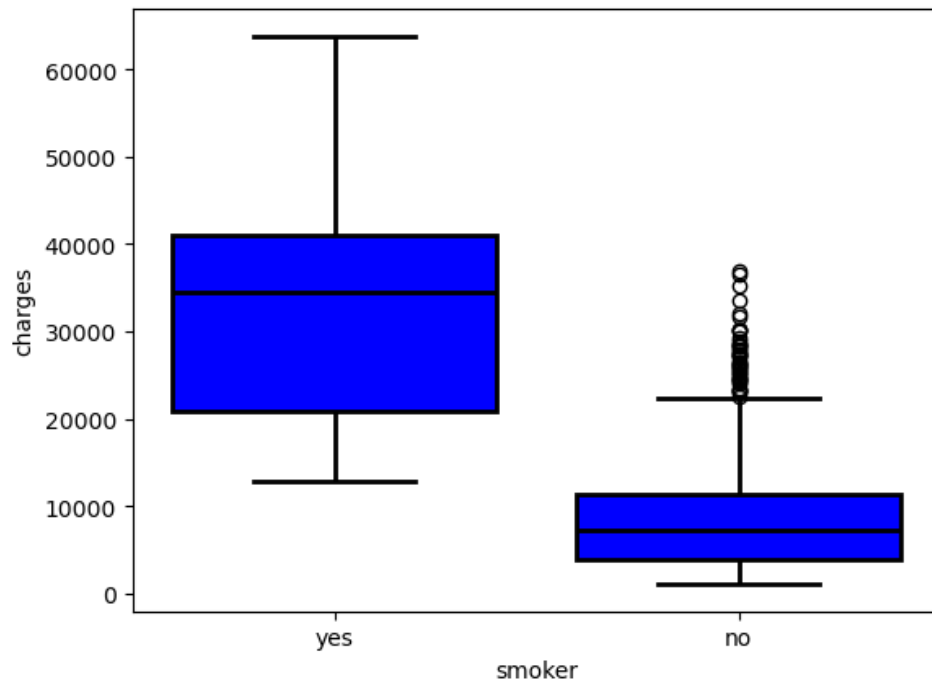


Figure 4. Distribution of charges for smokers versus non-smokers.

What it shows: two almost entirely separate distributions. Smokers' charges (median ≈\$34,500) sit well above non-smokers' (median ≈\$7,500), with the non-smoker group also showing several high-value outliers — individuals whose costs are elevated by some other factor, most plausibly high BMI, despite not smoking.

7.5 Figure 5 — Scatter Plot: BMI vs. Charges, by Smoking Status

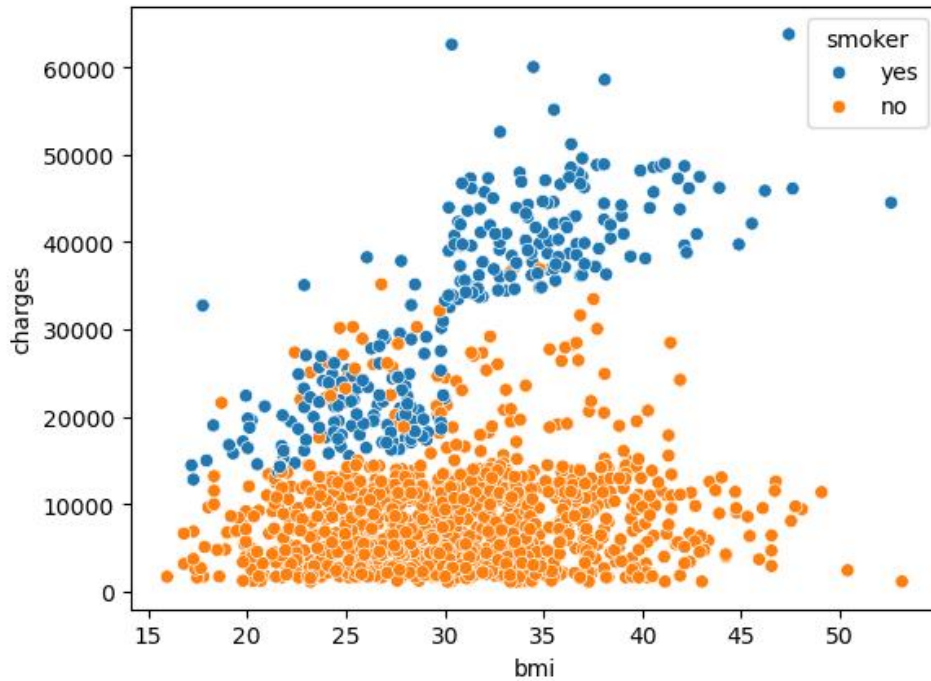


Figure 5. BMI plotted against charges, coloured by smoking status.

What it shows: the interaction effect described in Section 6.2. Below roughly BMI 30, smokers and non-smokers occupy overlapping cost ranges. Above BMI 30, smokers split into a visibly distinct, much higher band while non-smokers do not — this is the single most important chart in the report for explaining why a linear model underperforms Random Forest here.

7.6 Figure 6 — Pairplot: Age, BMI, and Charges by Smoking Status

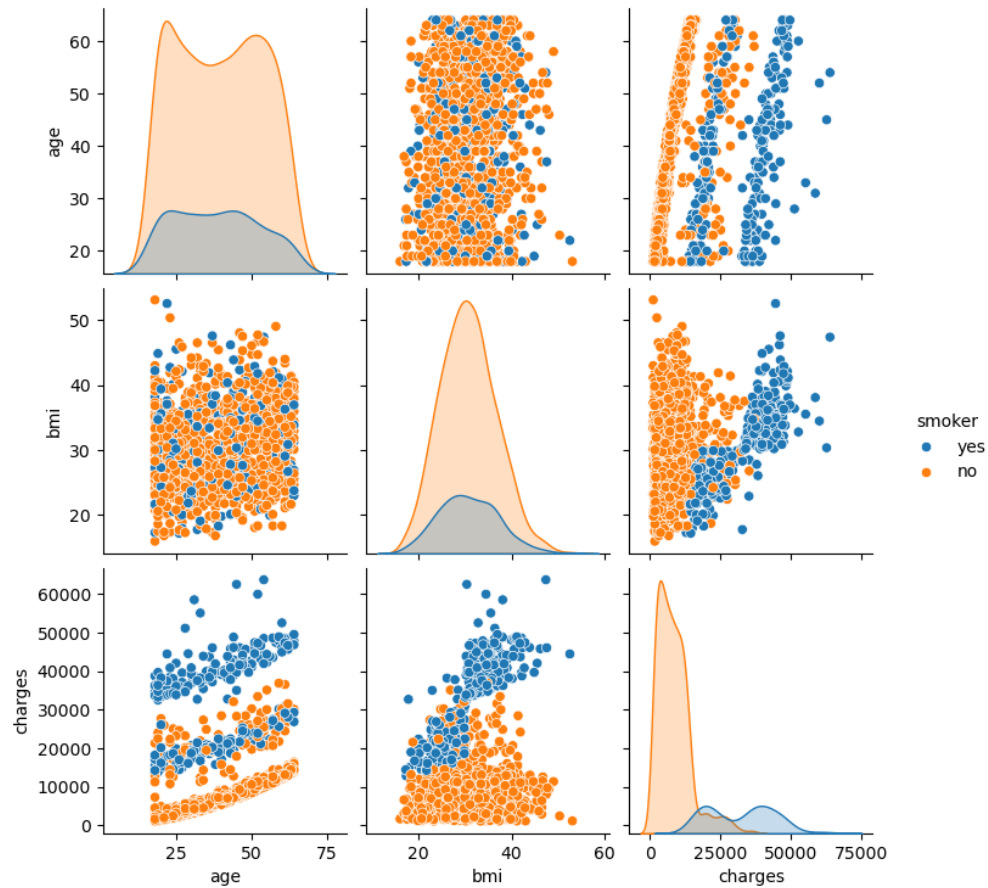


Figure 6. Pairwise relationships between age, bmi, and charges, split by smoking status.

What it shows: a consolidated view of every pairwise relationship at once. The age–charges panels show a steady upward drift within each smoking group; the bmi–charges panels reproduce the same divergence seen in Figure 5; and the diagonal charge distributions confirm that smokers’ charges are shifted well to the right of non-smokers’ across the board.

7.7 Figure 7 — Residual Plot

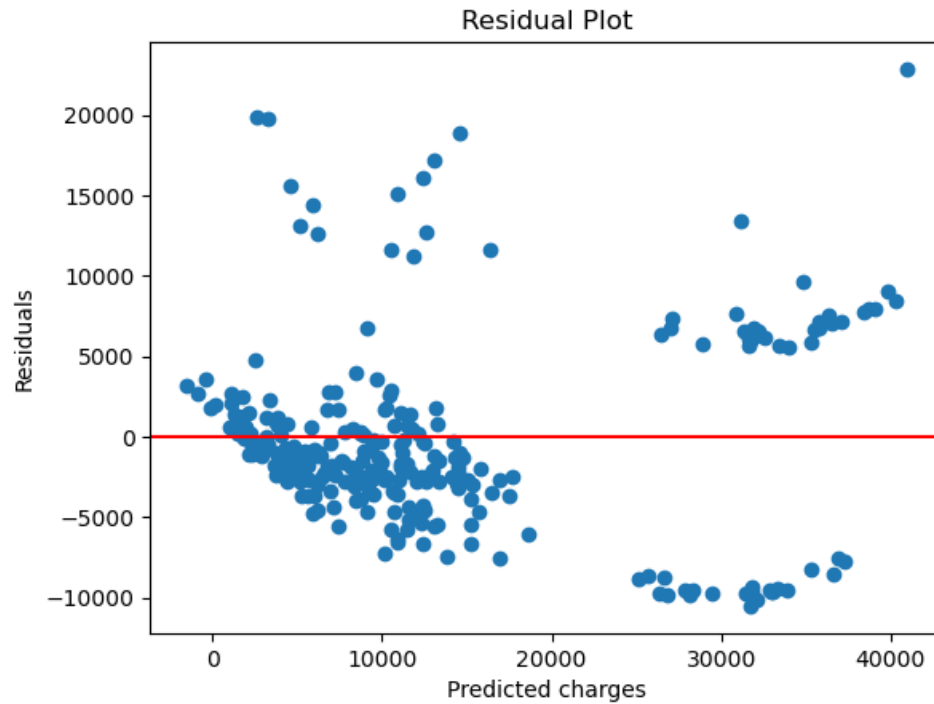


Figure 7. Residuals (actual – predicted charges) against predicted charges, linear regression.

What it shows: a clearly non-random pattern — residuals cluster into diagonal bands rather than scattering evenly around the red zero-line, and the spread widens noticeably at higher predicted values. This is the diagnostic evidence, referenced throughout Sections 5 and 6, that a single linear relationship cannot fully explain how charges are generated in this dataset.

8. Recommendations and Observations

8.1 Recommendations for Interpreting and Using This Model

1. Treat smoking status as the primary cost driver in any pricing or risk model.

At a correlation nearly three times stronger than the next-best predictor, smoking status should be the first variable any downstream pricing tier or risk segmentation is built around.

2. Do not price BMI and smoking status independently.

Because the two interact rather than simply adding together, a pricing model that scores "smoker" and "high BMI" as separate, additive risk factors will systematically underprice the highest-risk group (obese smokers) and overprice moderate-BMI smokers. A model or rate table that accounts for the combination explicitly will be more accurate than one that does not.

3. Use a non-linear model, or engineer an explicit interaction term, for production use.

Random Forest's advantage here is not incidental — it reflects a genuine structural feature of the data. If a linear model is preferred for interpretability, adding a single engineered feature ($\text{bmi} \times \text{smoker_e}$) would let a linear model capture much of this interaction directly, likely closing a meaningful part of the performance gap without needing a full ensemble model.

4. Report the cross-validated score, not the single best split.

The linear model's five-fold cross-validated mean (0.747) is a more honest estimate of real-world performance than the single test-split score (0.784), which happened to land on the favourable end of the observed range.

5. Treat sex, children, and region as low-priority factors for this dataset.

With correlations close to zero, these three variables added little value to the linear model here; a simplified model dropping them would likely perform almost identically, though they were retained in this analysis for completeness and comparability with the full feature set.

8.2 Technical Observations and Corrections

Several methodological points emerged during this project, in the same spirit as the technical notes documented for the earlier student project.

The RMSE variable in the model comparison loop is mislabeled.

As detailed in Section 5.3, the loop computes `mean_squared_error(y_test, y_pred)` and labels it `rmse` without taking the square root. The values printed (e.g. 33,596,915.85 for Linear Regression) are therefore MSE, not RMSE. The corrected line is:

```
from sklearn.metrics import root_mean_squared_error
rmse = root_mean_squared_error(y_test, y_pred)
```

Scaling was applied before the train/test split.

The same mild data-leakage pattern noted in the earlier project is present here: `StandardScaler` was fitted on the full dataset (including the eventual test rows) before the split. The more rigorous approach fits the scaler on training data only, ideally via a `Pipeline`, so that test data remains completely unseen until the final evaluation step.

`df2 = df` and `df4 = df3` create aliases, not independent copies.

Both assignments point to the same underlying object in memory rather than creating a separate copy. This did not cause an incorrect result in this notebook, but `.copy()` should be used wherever an independent, safely modifiable version of a `DataFrame` is intended.

`GridSearchCV` and `GradientBoostingRegressor` were imported but never used.

Both appear in the import cell but do not appear elsewhere in the notebook. Given that Random Forest's `max_depth=5` was set without testing alternatives, `GridSearchCV` would be a natural next step to confirm whether that hyperparameter choice is actually optimal.

8.3 Directions for Further Work

7. Add an explicit `smoker × bmi` interaction term (or a `smoker × bmi_bucket` categorical feature) to the linear models and re-evaluate — this is the single most promising next step given the finding in Section 6.2.
8. Apply a log-transformation to `charges` before modelling, given its strong right-skew, and compare whether this reduces the residual patterning observed in Figure 7.
9. Tune the Random Forest's hyperparameters (`max_depth`, `n_estimators`, `min_samples_leaf`) using `GridSearchCV` rather than the current fixed values.
10. Rebuild the pipeline using `sklearn`'s `Pipeline` class to eliminate the scaling-before-split leakage identified in Section 8.2.
11. Extract and report feature importances from the Random Forest model directly, to cross-check them against the linear correlation ranking in Section 5.1.

9. Conclusion

This project set out to determine which personal and lifestyle factors most strongly predict medical insurance charges, and how accurately those charges can be modelled using demographic and behavioural data alone.

The answer to the first question is unambiguous: **smoking status dominates every other factor**, correlating with charges at +0.79 — more than two and a half times the strength of the next predictor, age (+0.30). Critically, the analysis went further than a correlation ranking alone: the scatter plot in Figure 5 revealed that smoking status does not act independently of BMI, but **interacts** with it — obese smokers face disproportionately higher charges than either factor would predict on its own.

This interaction is the thread that ties the entire analysis together. It explains why the residual plot (Figure 7) showed clear, non-random banding rather than the scatter a well-fitted linear model should produce. It explains why **Random Forest ($R^2 = 0.873$) meaningfully outperformed Linear, Ridge, and Lasso regression (all ≈ 0.784 single-split, ≈ 0.747 cross-validated)** — tree-based models can represent conditional, non-additive relationships that a straight line structurally cannot. And it explains the pronounced right-skew in the charges distribution itself: the long tail of very high charges is disproportionately made up of exactly this subgroup.

This result is also a useful contrast with the earlier student-performance project, where the simplest linear model won precisely because the true relationship was linear. Together, the two projects demonstrate the same underlying lesson from opposite directions: **the right model is the one that matches the actual structure of the data**, and that structure has to be investigated — through correlation analysis, targeted visualisation, and residual diagnostics — rather than assumed in advance.

Methodologically, this project also reinforced the value of verifying a model rather than reporting a single score: the overfitting check, cross-validation, and residual analysis each added a layer of confidence (or, in the residual plot's case, a necessary caveat) that a bare R^2 figure alone would have missed entirely.

10. References and Appendices

10.1 Dataset

Medical Cost Personal Datasets (Insurance)

Published on Kaggle. Dataset of 1,338 individual insurance records covering demographic, lifestyle, and billing information.

Available at: <https://www.kaggle.com/datasets/mirichoi0218/insurance>

10.2 Statement of Independent Work

This project and its accompanying code were developed independently. No external notebook or third-party source code was referenced for this dataset; the analytical workflow reuses a general regression methodology previously developed and documented by the author on a separate dataset (student academic performance), applied here as independent practice on a new problem.

10.3 Library Documentation

Library	Documentation
pandas	https://pandas.pydata.org/docs/
NumPy	https://numpy.org/doc/
Matplotlib	https://matplotlib.org/stable/index.html
Seaborn	https://seaborn.pydata.org/
scikit-learn	https://scikit-learn.org/stable/

Specific API references:

- `pandas.get_dummies` — https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html
- `sklearn.model_selection.cross_val_score` — https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.cross_val_score.html
- `sklearn.ensemble.RandomForestRegressor` — <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
- `sklearn.metrics.root_mean_squared_error` — https://scikit-learn.org/stable/modules/generated/sklearn.metrics.root_mean_squared_error.html

Appendix A — Environment and Reproducibility

Item	Value
Language	Python 3
Notebook	insurance.ipynb (Jupyter)
Source file	insurance.csv
Records	1,338
Original columns	7
Features after encoding	8 (target excluded)
Train / test split	1,070 / 268 (80% / 20%)
Random seed	random_state=42

Appendix B — Final Feature Set (8 predictors)

#	Feature	Origin
1	age	Original numeric
2	bmi	Original numeric
3	children	Original numeric
4	sex_e	Binary encoding of sex
5	smoker_e	Binary encoding of smoker
6	region_northwest	One-hot from region
7	region_southeast	One-hot from region
8	region_southwest	One-hot from region

Target: charges (continuous)

Appendix C — Model Results Summary

Model	Hyperparameters	R ² (test set)	RMSE (corrected)
Linear Regression	default	0.784	\$5,796.28
Ridge	alpha=1.0	0.784	\$5,796.98
Lasso	alpha=0.1	0.784	\$5,796.32

Model	Hyperparameters	R ² (test set)	RMSE (corrected)
Random Forest	n_estimators=100, max_depth=5	0.873	\$4,436.24

Cross-validated Linear Regression (5-fold): mean R² = 0.747, range 0.706–0.778. Best performing single model: Random Forest (R² = 0.873).

Appendix D — DataFrame Lineage

Object	Derivation	Purpose
df	pd.read_csv(...)	Raw data, no missing values
df2	df (alias)	Working reference; binary encodings added
df3	pd.concat([df2, region_dummies], axis=1) then text columns dropped	Fully numeric, model-ready
df4	df3 (alias)	Working reference for feature/target split
x	df4.drop('charges', axis=1)	8 predictor features
y	df4['charges']	Target variable
x_scaled	StandardScaler().fit_transform(x)	Standardised features